Table of Contents

2. Abstract

Rapid industrialization and unregulated discharge of untreated effluents have made heavy metal contamination one of the most pressing environmental concerns in India. Industries such as tanneries, textile units, electroplating plants, and mining operations release toxic metal ions - including lead (Pb^{2+}), chromium (Cr^{3+}/Cr^{6+}), mercury (Hg^{2+}), and copper (Cu^{2+}) - into water bodies, causing severe ecological and public health consequences [1].

The Central Water Commission, in its 2024 monitoring assessment, examined 688 sites across major Indian rivers and recorded heavy metal concentrations exceeding permissible limits at numerous stations, particularly along the Kanpur–Varanasi stretch of the Ganga River, where tannery effluents remain the dominant source of chromium loading [2].

As per the Central Pollution Control Board (CPCB), 37 river stretches across India were classified as critically polluted in 2023, with industrial heavy metal discharge identified as a principal causative factor [3]. Conventional treatment techniques, including chemical precipitation, coagulation-flocculation, and biological oxidation, have demonstrated limited selectivity and precision for trace-level heavy metal removal from complex effluent matrices, warranting the exploration of more chemically directed analytical and remediation strategies [4].

The present study investigates the application of a complexometric approach - employing chelating ligands such as ethylenediaminetetraacetic acid (EDTA), dithiocarbamate (DTC), and thiourea - for both analytical quantification and physicochemical treatment of heavy metal-contaminated wastewater.

Complexometric titration using EDTA as the primary chelating agent provides a cost-effective and analytically reproducible method for quantitative estimation of Pb^{2+}, Mn^{2+}, Hg^{2+}, and Cu^{2+} in industrial effluent samples, employing metallochromic indicators such as Xylenol Orange, PAN (1-(2-pyridylazo)-2-naphthol), and Dithizone for precise endpoint detection [5].

The findings establish that the complexometric approach provides a reliable, field-adaptable, and low-cost alternative to instrument-intensive techniques for routine heavy metal monitoring, particularly in resource-constrained settings prevalent across Indian industrial clusters. The study holds significance in the context of national initiatives such as the Namami Gange Mission and India's National Water Quality Monitoring Programme, both of which prioritize heavy metal remediation as an immediate environmental objective [2], [3].

CHAPTER 1

Introduction to Wastewater Treatment

1.1 Background and Global Water Pollution Scenario

Water is among the most fundamental natural resources sustaining life on Earth, yet its degradation through anthropogenic activities has emerged as one of the defining environmental challenges of the twenty-first century. The worldwide demand for fresh water is growing at an estimated rate of one percent annually, and projections indicate a twenty to thirty percent increase in total global water consumption by 2050 [6].

Parallel to this escalating demand, industrial expansion, rapid urbanization, and intensified agricultural activity have led to the continuous discharge of inadequately treated effluents into surface water bodies, groundwater aquifers, and coastal ecosystems. Pollutants introduced through these pathways include heavy metals, synthetic organic compounds, nutrients, and pathogens, each of which presents distinct hazards to aquatic ecosystems and human health [7].

In the Indian context, water pollution has attained a critical dimension owing to the scale and diversity of industrial activity, the dense population dependent on river systems for domestic and agricultural use, and persistent gaps in wastewater management infrastructure. India accounts for approximately seventeen percent of the world's scattered municipal waste, and an estimated seventy-two percent of wastewater generated in urban India is discharged without adequate treatment into receiving water bodies [8].

As per data compiled by the Central Pollution Control Board (CPCB) under the Ministry of Environment, Forest and Climate Change (MoEFCC), approximately thirty-nine percent of sewage treatment plants (STPs) across the country fail to meet the effluent quality standards prescribed under the Environment Protection Rules, 1986 [9].

The National Water Quality Monitoring Network of CPCB, comprising over 4,736 monitoring sites across rivers, lakes, canals, and drains, has systematically documented the deterioration of surface water quality across twenty-eight states, with heavy metal contamination identified as a priority concern at a significant proportion of monitored stations [3].

1.2 Sources and Types of Wastewater in India

Wastewater in India originates from three principal categories of sources: industrial effluents, domestic or municipal sewage, and agricultural runoff [10].Each category differs substantially in its physicochemical composition, volume, and degree of treatability.

Industrial Effluents: Industrial discharge constitutes one of the most chemically complex and hazardous streams of wastewater generated in India. Sectors such as tanning, textile dyeing, electroplating, metal processing, mining, and chemical manufacturing collectively release effluents containing heavy metals, synthetic dyes, organic solvents, acids, and alkalis [7].

Of particular concern are the tannery clusters concentrated in cities such as Kanpur (Uttar Pradesh), Chennai (Tamil Nadu), and Kolkata (West Bengal), which discharge chromium-rich effluents, sulfides, and high biological oxygen demand (BOD) wastes into proximate river systems. The Ganga basin stretch from Kannauj to Kanpur to Varanasi has been identified as among the most critically polluted stretches in the country, with analysis of upstream and downstream water and sediment samples revealing ten-fold increases in chromium concentration at tannery discharge points [11]. As per CPCB data from 2017–18 to 2021–22, Kanpur's tannery sector alone ranked as the largest industrial contributor of hazardous waste volumes in the Ganga basin region [12].

Domestic and Municipal Sewage: Urban India generates approximately 72,368 million litres per day (MLD) of domestic wastewater, of which only a fraction receives adequate treatment before discharge [9]. Population growth, rapid urbanization, and insufficient investment in sewerage infrastructure have widened the gap between sewage generation and treatment capacity. Namami Gange Programme has sanctioned 161 sewage management projects worth over ₹24,581 crore to create and rehabilitate 5,501 MLD of sewage treatment capacity, yet as of 2022, only approximately thirty-two percent of this sanctioned capacity had been operationalized [8].

Agricultural Runoff: Non-point source pollution from agricultural activities contributes significantly to water quality degradation, particularly in rural and semi-urban areas. Excessive use of chemical fertilizers, pesticides, and irrigation water leads to the leaching of nitrates, phosphates, and trace heavy metals such as arsenic, lead, and cadmium into both surface and groundwater bodies [10]. The contamination of groundwater in Haryana, Punjab, and Uttar Pradesh with heavy metals and nitrates has been directly linked to intensive agricultural and associated small-scale industrial activity in these regions [13].

1.3 Conventional Methods of Wastewater Treatment and their Limitations

Wastewater treatment has traditionally been organized into three sequential stages primary, secondary, and tertiary treatment each targeting a distinct class of pollutants through physical, biological, and chemical mechanisms respectively [14].

Primary Treatment involves physical processes such as screening, sedimentation, and skimming to remove suspended solids, floating materials, and settleable matter. Effective for gross particulate removal, primary treatment demonstrates negligible efficiency in eliminating dissolved heavy metal ions, which remain fully in solution and pass through sedimentation units unaffected [15].

Secondary Treatment employs biological processes - principally activated sludge, trickling filters, and rotating biological contactors - to degrade dissolved organic matter through microbial activity. Aerobic treatment is generally suited to low- and medium-strength effluents, while anaerobic systems are deployed for high-strength industrial wastewaters. However, conventional biological treatment is not designed for the removal of heavy metals, and high metal concentrations in influent streams can in fact inhibit or toxify the microbial biomass responsible for organic degradation, further compromising treatment performance [16].

Tertiary or Advanced Treatment encompasses chemical precipitation, coagulation–flocculation, ion exchange, membrane filtration, and adsorption. Chemical precipitation using lime or sodium hydroxide can remove certain metal ions by converting them to insoluble hydroxides; however, this approach is non-selective, generates large volumes of metal-laden sludge requiring further treatment, and is ineffective at trace-level metal concentrations where solubility equilibria may favour metal retention in solution [7]. Coagulation–flocculation, while cost-effective for suspended solids removal, involves high chemical consumption and exhibits variable performance depending on influent pH and competing ionic species [17]. Ion exchange and membrane-based techniques such as reverse osmosis offer higher metal removal efficiencies but are capital-intensive, prone to fouling, and operationally demanding - constraints that limit their deployment at the scale and frequency required across India's diverse industrial and municipal effluent generating sectors [14].

The limitations of these conventional approaches - particularly their inability to selectively and quantitatively remove trace heavy metal ions from complex effluent matrices - underscore the necessity for more chemically informed and analytically precise treatment and monitoring strategies, such as the complexometric approach explored in the present study [7], [15].

1.4 Need for a Complexometric Approach

Complexometric methods, rooted in coordination chemistry, offer a fundamentally different analytical and remediation framework compared to conventional physicochemical treatment. The core principle involves the formation of stable, soluble metal–ligand complexes through the donation of electron pairs from a chelating agent such as ethylenediaminetetraacetic acid (EDTA), dithiocarbamate, or thiourea to a metal ion centre [5].

The thermodynamic stability of these complexes, quantified through formation or stability constants, governs the selectivity and completeness of metal chelation under defined pH and ionic conditions [6].

From an analytical standpoint, complexometric titration using EDTA as the chelating agent provides a precise, reproducible, and cost-accessible method for the quantitative determination of metal ion concentrations in water and wastewater samples. The technique is particularly well-suited to the resource constraints prevalent in Indian environmental testing laboratories, where instrument-intensive methods such as inductively coupled plasma–mass spectrometry (ICP-MS) or graphite furnace atomic absorption spectrometry (GFAAS) may not be routinely available [5].

The application of selective metallochromic indicators - including Xylenol Orange for Pb^{2+}, PAN (1-(2-pyridylazo)-2-naphthol) for Mn^{2+}, and Dithizone for Hg^{2+} - permits endpoint-specific detection with sufficient sensitivity for regulatory monitoring purposes [5].

Beyond analytical quantification, complexometric principles have also been extended to wastewater treatment itself, through the development of EDTA-functionalized adsorbents, chelating resins, and hybrid membrane-complexation systems capable of achieving selective and regenerable heavy metal removal from industrial effluents [4].

This dual role as both an analytical tool and a treatment strategy distinguishes the complexometric approach as a uniquely versatile framework for addressing heavy metal contamination in wastewater, and forms the conceptual foundation of the present investigation.

CHAPTER 2

Heavy Metal Chemistry in Aqueous Systems

2.1 Definition and Classification of Heavy Metals

The term "heavy metal" broadly refers to metallic elements possessing relatively high atomic masses and densities at least five times greater than that of water (5 g/cm^3) [18]. This group encompasses a diverse array of elements including lead (Pb), chromium (Cr), mercury (Hg), copper (Cu), cadmium (Cd), zinc (Zn), manganese (Mn), arsenic (As), and nickel (Ni), among others. Based on their biological and environmental significance, heavy metals are typically categorized into two broad groups: essential trace metals and non-essential toxic metals [19].

Essential metals such as copper, zinc, and manganese serve critical biochemical roles in enzymatic catalysis, electron transport, and hormonal regulation; however, they become toxic when present in concentrations exceeding physiological thresholds. Non-essential metals such as lead, mercury, and cadmium, by contrast, have no recognized beneficial biological function and exert toxicity even at trace concentrations through interference with enzyme function, displacement of essential metal cofactors, and induction of oxidative stress [20]. The toxicological significance of a heavy metal in an aqueous environment is determined not merely by its total concentration but by its chemical speciation that is, the physicochemical form in which it exists in solution. Metal speciation governs bioavailability, mobility, and reactivity [21].

For example, chromium exists in two principal oxidation states in aquatic systems: the trivalent form Cr(III), which is relatively less toxic and tends to precipitate as $Cr(OH)_3$ at neutral to alkaline pH, and the hexavalent form Cr(VI), which is highly soluble, highly mobile, and classified as a Group I human carcinogen by the International Agency for Research on Cancer (IARC) [22].

Similarly, inorganic mercury Hg^{2+} in aquatic systems undergoes microbially mediated methylation to form methylmercury (CH_3Hg^+), a far more bioaccumulative and neurotoxic species capable of biomagnifying through aquatic food chains [23].

Understanding the speciation of heavy metals in the aqueous phase is therefore a prerequisite for accurate environmental risk assessment and for the design of selective remediation approaches a principle that underlies the complexometric methodology explored in this study.

2.2 Sources of Heavy Metal Contamination in Indian Water Bodies

Heavy metal contamination of Indian surface water and groundwater arises from a combination of geogenic and anthropogenic sources, with the latter dominating due to the scale and diversity of industrial activity across the country [19].

2.2.1 Industrial Sources

Industrial discharge constitutes the most concentrated and chemically complex source of heavy metal inputs into Indian water bodies. Tanneries represent the single largest industrial contributor of chromium contamination, particularly in the Kanpur–Jajmau cluster of Uttar Pradesh, where chromium-rich effluents from chrome tanning operations have been documented to increase chromium concentrations ten-fold between upstream and downstream sediment sampling points along the Ganga River [11].

The leather industry accounts for the livelihoods of millions of workers across India and contributes significantly to national export revenues; however, an estimated 80–90% of tanneries employ chromium as the primary tanning agent, of which the hides absorb only 50–70%, with the remainder entering wastewater streams [24].

Indian discharge norms under Schedule VI of the Environment Protection Rules, 1986 set the permissible total chromium limit at 2.0 mg/L and hexavalent chromium at 0.1 mg/L for inland surface water discharge; yet monitoring data consistently reveal violations, particularly in smaller cluster-based units lacking adequate effluent treatment infrastructure [9].

The textile and dyeing industry, concentrated in cities such as Surat (Gujarat), Ludhiana (Punjab), Tiruppur (Tamil Nadu), and Bhilwara (Rajasthan), discharges heavy metals including copper, chromium, and zinc as by-products of dyeing and finishing processes, alongside high BOD and COD loads from synthetic dyes [10].

Electroplating industries are significant sources of cadmium, nickel, and copper contamination in peri-urban industrial zones. Mining operations in mineral-rich states such as Jharkhand, Odisha, Chhattisgarh, and Rajasthan release arsenic, lead, manganese, and iron into proximate river systems through acid mine drainage and tailings runoff [25].

2.2.2 Agricultural Sources

The extensive application of chemical fertilizers, pesticides, and irrigation with untreated wastewater in Indian agriculture contributes significantly to diffuse heavy metal loading of both surface water and groundwater [10].

Phosphate fertilizers are known to introduce cadmium and arsenic into agricultural soils, from which leaching into groundwater occurs progressively under monsoon recharge conditions.

States including Punjab, Haryana, Uttar Pradesh, and Bihar have documented elevated arsenic and lead contamination of shallow groundwater aquifers attributable in part to agricultural chemical use [13].

2.2.3 Municipal and Domestic Sources

Urban sewage and municipal solid waste leachate contribute copper, zinc, lead, and manganese to receiving water bodies through poorly controlled discharge. The corroding infrastructure of aging water supply pipelines - many of which in older Indian cities still incorporate lead soldering joints - additionally introduces Pb^{2+} into distributed drinking water at the point of use [20]. As per the CPCB's 2023 river quality assessment, groundwater in as many as 153 Indian districts has been found contaminated with arsenic, 93 districts with lead, 30 districts with chromium, and 24 districts with cadmium, reflecting the pervasive and geographically widespread nature of heavy metal contamination across the country [25].

2.3 Chemistry of Specific Heavy Metals in Aqueous Medium

A detailed understanding of the aqueous chemistry of each metal ion addressed in the present study is essential for interpreting their complexometric behaviour and environmental fate.

2.3.1 Lead (Pb^{2+})

Lead enters aquatic systems predominantly in the divalent form Pb^{2+}, which exhibits moderate aqueous solubility and a tendency to form stable complexes with sulfate, carbonate, hydroxide, and organic ligands [21]. In the pH range of natural waters (6.5–8.5), Pb^{2+} can precipitate as $PbCO_3$ (cerussite) or $Pb(OH)_2$ at higher pH values; however, in acidic industrial effluents it remains predominantly in the free ionic form, making complexometric estimation feasible.

Lead demonstrates a high affinity for sulfur-donor ligands - a property exploited by dithiocarbamate chelation, which forms highly stable Pb–DTC complexes through S,S-coordination in aqueous medium [26].

2.3.2 Chromium (Cr^{3+} / Cr^{6+})

Chromium presents a unique dual-speciation behaviour in aqueous systems. Trivalent chromium Cr(III), the dominant form in tannery effluents, forms stable hexaaqua complexes $[Cr(H_2O)_6]^{3+}$ in acidic solutions, which hydrolyze progressively at increasing pH to form $Cr(OH)^{2+}$, $Cr(OH)_2^{+}$, and ultimately insoluble $Cr(OH)_3$ precipitate [22]. Hexavalent chromium Cr(VI), which can form under oxidizing conditions through the oxidation of Cr(III), exists as chromate (CrO_4^{2-}) at alkaline pH and dichromate ($Cr_2O_7^{2-}$) at acidic pH - both highly soluble and mobile anionic species that penetrate cellular membranes and cause oxidative DNA damage [27].

The Sukinda Valley of Odisha, which hosts one of the world's largest chromite ore deposits, has ranked among most chromium-polluted regions globally, with surface water containing Cr(VI) concentrations exceeding the national standard by more than double at numerous sampling points [22]. EDTA forms a thermodynamically stable 1:1 chelate complex with Cr^{3+}, making it analytically suitable for chromium determination via back-titration methods [5].

2.3.3 Mercury (Hg^{2+})

Mercury is classified by the Agency for Toxic Substances and Disease Registry (ATSDR) as the third most hazardous substance globally, ranking behind arsenic and lead [23]. Inorganic mercury Hg^{2+} enters Indian water systems primarily from chloralkali plants, paper and pulp manufacturing, paint industries, and pharmaceutical and medical device production [19]. In aqueous environments, aquatic microorganisms - particularly sulfate-reducing bacteria - convert Hg^{2+} to methylmercury (CH_3Hg^+) through anaerobic methylation, a process that dramatically increases mercury's bioaccumulation potential and neurotoxicity. Studies from the East Kolkata Wetlands, a Ramsar site receiving approximately 600 million litres of composite industrial effluent per day, have documented the presence of mercury and lead in fish tissues at concentrations posing non-negligible human health risks through dietary exposure [28]. Mercury forms highly stable complexes with thiourea through coordination via the sulfur atom, a property that underpins both its analytical determination and the remediation strategies discussed in Chapter 6 [5].

2.3.4 Copper (Cu^{2+})

Copper is an essential micronutrient at low concentrations (recommended daily intake 0.9 mg), serving as a cofactor for numerous oxidoreductase enzymes; however, excess copper induces oxidative stress through generation of reactive oxygen species (ROS) and has been linked to liver damage, gastrointestinal disorders, and, with chronic exposure, neurological disturbances [23].

In industrial wastewater, Cu^{2+} originates primarily from electroplating baths, mining operations, wood pulp production, and copper-based fungicide use in agriculture. The acceptable limit for copper in drinking water under BIS IS:10500:2012 is 0.05 mg/L. Copper's strong affinity for dithiocarbamate ligands through N,N-disubstituted coordination renders Cu–DTC complexes amenable to spectroscopic characterization and quantitative analytical estimation [26].

2.3.5 Manganese (Mn^{2+})

Manganese is a naturally occurring element that enters water bodies through geogenic dissolution of manganese-containing minerals, mining discharge, and industrial effluents from dry-cell battery and fertilizer manufacturing. While it is an essential dietary element at trace concentrations, chronic exposure to elevated Mn^{2+} in drinking water has been associated with neurotoxic effects - particularly in children - manifesting as cognitive and behavioural deficits resembling Parkinson's disease pathology [23].

2.4 Health and Ecological Impacts of Heavy Metal Contamination

The persistence, non-biodegradability, and bioaccumulative properties of heavy metals distinguish them from other classes of water pollutants and render their ecological and human health consequences particularly severe [20].

Human Health Impacts: Lead exposure at elevated concentrations causes hypertension, cognitive impairment, neurodevelopmental deficits in children, and disruption of renal and haematopoietic function in adults [29]. A human health risk assessment study conducted across seven sampling locations in Maharashtra found that arsenic contributed the most to ingestion-based health risk, while mercury and lead concentrations in certain samples posed non-carcinogenic hazards that warranted immediate remediation intervention [29]. Hexavalent chromium is a recognized respiratory carcinogen and causes DNA strand breaks, chromosomal aberrations, and disruption of cellular redox homeostasis upon inhalation or ingestion [27]. Mercury's neurotoxicity, famously documented in the Minamata disease outbreak in Japan, extends to Indian populations exposed through contaminated fish consumption and occupational handling; symptoms include cognitive dysfunction, sensory impairment, hearing and visual disorders, and renal damage [23]. In India, approximately 37.7 million individuals are affected by waterborne illnesses annually, and heavy metal contamination has been implicated as a contributing factor to the disease burden in industrial regions of Maharashtra, Uttar Pradesh, Tamil Nadu, and West Bengal [29].

Ecological Impacts: At the ecosystem level, heavy metals disrupt aquatic biodiversity by inhibiting enzymatic activity in aquatic organisms, impairing reproduction, and accumulating through food chains via the process of biomagnification [20]. Fish kills, decline of macroinvertebrate populations, and sediment toxicity have been documented in river stretches downstream of tannery and electroplating effluent discharge points across India. The East Kolkata Wetlands study demonstrated elevated concentrations of chromium, vanadium, lead, and mercury in fish muscle, gill, and kidney tissues at concentrations exceeding safe dietary exposure thresholds [28]. Furthermore, heavy metals adsorbed onto suspended sediments can be transported long distances downstream before being remobilized under changing pH and redox conditions, exacerbating the spatial extent of contamination beyond the immediate discharge zone [19].

2.5 Regulatory Standards for Heavy Metals in India

India's regulatory framework for heavy metal limits in water is established through three primary instruments: the Bureau of Indian Standards specification IS:10500:2012 for drinking water, CPCB effluent discharge standards under Schedule VI of the Environment Protection Rules, 1986, and WHO drinking water quality guidelines [9]. Table 2.1 summarizes the permissible limits applicable to the key metals addressed in this study.

Table 2.1: Permissible Limits for Key Heavy Metals in Water — BIS, CPCB, and WHO

Heavy Metal	BIS IS:10500:2012 (Drinking Water, mg/L)	CPCB Discharge Limit (Inland Water, mg/L)	WHO Guideline (mg/L)
Lead (Pb)	0.01	0.1	0.01
Chromium Total (Cr)	0.05	2.0	0.05
Chromium Cr(VI)	0.05	0.1	0.05
Mercury (Hg)	0.001	0.01	0.006
Copper (Cu)	0.05	3.0	2.0
Manganese (Mn)	0.1	2.0	0.4

Sources: BIS IS:10500:2012 [9]; CPCB Schedule VI, Environment Protection Rules 1986 [9]; WHO Guidelines for Drinking Water Quality, 4th Ed. [30]

CHAPTER 3

Ligands and Their Complex Formation Behaviour

3.1 Introduction to Ligands and Coordination Chemistry

The chemistry of coordination compounds - complexes formed between a central metal ion and surrounding molecules or ions called ligands - constitutes one of the most fundamental and applied branches of inorganic and analytical chemistry [31]. A ligand is defined as any atom, ion, or molecule that possesses at least one lone pair of electrons and is capable of donating these electrons to a metal ion through the formation of a coordinate covalent bond, thereby acting as a Lewis base to the Lewis acidic metal centre [32]. The resulting coordination compound, or complex, is stabilized through a combination of electrostatic attraction, orbital overlap, and the geometric constraints imposed by the ligand structure.

In the context of wastewater treatment and environmental analytical chemistry, the selection of an appropriate ligand is critical. The ligand must fulfil several criteria simultaneously: it must form stable and selective complexes with target metal ions at environmentally relevant pH values; it must be readily available and chemically reproducible in laboratory or field conditions; and its complexes must yield analytical signals - colour changes, solubility changes, or spectroscopic transitions - that are practically detectable [31]. The three ligands central to this investigation - ethylenediaminetetraacetic acid (EDTA), dithiocarbamate (DTC), and thiourea - each satisfy these requirements for specific metal ion systems encountered in Indian industrial wastewater, namely Pb^{2+}, Cr^{3+}, Hg^{2+}, and Cu^{2+}.

3.2 Classification of Ligands Based on Denticity

The most fundamental classification of ligands is based on their **denticity** - the number of donor atoms through which a single ligand molecule coordinates to a metal ion [32]. This classification directly governs the stability and analytical utility of the resulting complex.

Monodentate ligands possess a single donor atom and form only one coordinate bond with the metal centre. Common examples include water (H_2O), ammonia (NH_3), chloride (Cl^-), and cyanide (CN^-). Complexes formed with monodentate ligands tend to be kinetically labile and thermodynamically less stable, as the ligand can be readily displaced by competing species in solution. In wastewater matrices, where multiple competing metal ions and anions are present, monodentate ligands lack the selectivity required for precise analytical work [33].

Bidentate ligands contain two donor atoms and form two simultaneous coordinate bonds with the metal, generating a stable five- or six-membered chelate ring. The formation of a chelate ring - a cyclic structure in which the metal and ligand together form a closed ring - dramatically increases complex stability through the well-established **chelate effect** [34]. Examples of bidentate ligands include ethylenediamine (en, an N,N-donor), oxalic acid (an O,O-donor), and dithiocarbamate (DTC, an S,S-donor). Both dithiocarbamate and thiourea - two of the ligands employed in this study - belong to this bidentate category, coordinating through sulfur donor atoms with high affinity for soft metal ions such as Pb^{2+}, Hg^{2+}, and Cu^{2+} [33].

Polydentate and hexadentate ligands possess three or more donor atoms and form multiple simultaneous coordinate bonds with a single metal centre. Among these, EDTA is the most important, functioning as a hexadentate (six-donor) ligand capable of forming five stable chelate rings with a single metal ion in a single complexation reaction. This exceptional denticity makes EDTA uniquely effective as a chelating agent for a broad range of divalent and trivalent metal ions across varying pH conditions [31]. Tridentate ligands such as diethylenetriamine (dien) and macrocyclic ligands such as crown ethers also belong to this broader polydentate category and are used in more specialized separation and sensing applications.

3.3 Ethylenediaminetetraacetic Acid (EDTA) - Structure, Properties and Chelation Chemistry

Ethylenediaminetetraacetic acid (EDTA), with the molecular formula $C_{10}H_{16}N_2O_8$ and IUPAC name 2,2′,2″,2‴-(ethane-1,2-diyldinitrilo)tetraacetic acid, is the most widely employed chelating agent in analytical and industrial chemistry [31]. EDTA is a tetraprotic acid (H_4Y) that undergoes progressive dissociation across four steps as pH is increased, yielding the ionic species H_3Y^-, H_2Y^{2-}, HY^{3-}, and ultimately the fully deprotonated Y^{4-} anion at pH values above approximately 10.2 [34]. It is this fully deprotonated tetraanionic form Y^{4-} that acts as the hexadentate chelating agent, donating electron pairs from two tertiary nitrogen atoms and four carboxylate oxygen atoms simultaneously to a central metal ion.

3.3.1 Structural Features

The EDTA molecule comprises a central ethylene bridge ($–CH_2–CH_2–$) connecting two nitrogen atoms, each of which bears two acetic acid ($–CH_2COOH$) pendant arms. This architecture creates six donor sites arranged in a manner that can wrap around a metal ion to form five stable five-membered chelate rings simultaneously - one ring involving the N–C–C–N backbone and four additional rings each formed by an N–C–C–O carboxylate arm [31]. The resulting $[M(EDTA)]^{n-}$ complex adopts a distorted octahedral geometry, with bond angles deviating from the ideal 90° of a true octahedron due to the geometric constraints imposed by the ethylene backbone - a feature documented through resonant inelastic X-ray scattering studies of Fe(III)-EDTA complexes [35].

3.3.2 pH Dependency of EDTA Complexation

A critical feature of EDTA coordination chemistry is its strong pH dependence. Because the fully effective chelating species is Y^{4-}, which is generated only at alkaline pH, the conditional stability constant K'f - representing the effective stability of the metal-EDTA complex under actual pH conditions - decreases substantially as the solution pH decreases below 10. This is quantified through the alpha (α) factor, which represents the fraction of total EDTA present as the free Y^{4-} species at any given pH [34]. For analytical purposes, EDTA titrations are typically performed at pH 10 using an ammonia–ammonium chloride buffer for the determination of Ca^{2+}, Mg^{2+}, and total water hardness, while other pH values are used for selective determination of specific metal ions. For Pb^{2+} determination, pH 5–6 in an acetic acid–sodium acetate buffer is recommended; for Cr^{3+}, near-neutral to mildly alkaline conditions are preferred [26].

3.3.3 Stability Constants of Metal-EDTA Complexes

The formation or stability constant (Kf, commonly expressed as log Kf) quantifies the thermodynamic tendency of a metal ion to form a complex with EDTA. A higher log Kf value indicates a more stable complex and therefore greater analytical sensitivity and completeness of reaction at the equivalence point [34]. Table 3.2 lists log Kf values for the key metal-EDTA complexes relevant to this study.

3.4 Dithiocarbamate (DTC) Ligands - Structure, Properties and Metal Affinity

Dithiocarbamate (DTC) ligands represent a large and chemically diverse family of organosulfur compounds with the general formula $R_2NCS_2^-$, derived from the reaction of secondary amines (R_2NH) with carbon disulfide (CS_2) in the presence of a base such as sodium hydroxide [33]. The resulting dithiocarbamate anion possesses two sulfur donor atoms connected through a thiocarbonyl moiety (–C(=S)–S^-), which can coordinate to a metal centre in either monodentate or bidentate fashion, with bidentate S,S-coordination being overwhelmingly preferred due to the enhanced stability conferred by the chelate effect [33].

3.4.1 Electronic Structure and Bonding

The dithiocarbamate ligand is characterized by a pronounced resonance between two principal electronic forms: the thioureide form, in which the C–N bond has partial double bond character (represented as –N^+=C–S^-/–S^-), and a form in which negative charge is localized symmetrically across both sulfur atoms [33]. This delocalization of electron density through N-to-C π bonding results in two key structural consequences: (i) enhanced basicity of the sulfur donor atoms compared to related ligands such as xanthates and dithiocarboxylates, which translates to stronger metal-sulfur bonds; and

(ii) a high rotational barrier around the C–N bond, which restricts conformational flexibility and imposes a relatively rigid planar geometry on the coordinating S–C–S unit [33].

According to the Hard and Soft Acids and Bases (HSAB) principle formulated by Pearson, the sulfur donor atoms of dithiocarbamate are classified as **soft bases**, which preferentially interact with **soft acid** metal ions such as Hg^{2+}, Pb^{2+}, Cd^{2+}, and Cu^{2+} - precisely the metal ions of concern in Indian tannery, electroplating, and chloralkali industrial effluents [31].

This HSAB-guided selectivity is one of the principal reasons for choosing dithiocarbamate in the present study for the selective chelation of Pb^{2+} and Cu^{2+} from complex industrial wastewater matrices.

3.4.2 Synthesis of Dithiocarbamate Ligands

Dithiocarbamate salts are readily synthesized through a straightforward and atom-efficient reaction between a secondary amine, carbon disulfide, and sodium hydroxide:

$R_2NH + CS_2 + NaOH \rightarrow R_2NCS_2Na + H_2O$

This reaction proceeds quantitatively at room temperature in aqueous or alcoholic solvents, yielding the sodium dithiocarbamate salt as a stable, crystalline product [33]. The simplicity and cost-effectiveness of this synthesis render dithiocarbamate ligands particularly attractive for environmental analytical applications, including those envisaged for use in Indian industrial monitoring contexts.

3.4.3 Coordination Modes and Complex Geometry

When coordinating to metal ions in bidentate mode, the DTC ligand forms a three-membered M–S–C–S–M chelate ring, which, while smaller than the five-membered rings typical of EDTA complexes, is stabilized by the planarity of the S–C–S unit and the strong covalent character of metal-sulfur bonds [33]. For divalent metal ions such as Pb^{2+} and Cu^{2+}, which typically adopt four-coordinate environments, two bidentate DTC ligands coordinate to form neutral $[M(DTC)_2]$ complexes of square planar or tetrahedral geometry, as confirmed by FTIR spectral data showing characteristic C–S stretching bands in the range 1012–1078 cm^{-1} and C=N thiouride bands in the range 1425–1499 cm^{-1} [33]. The IR band splitting pattern - specifically, whether one or two bands appear in the 950–1050 cm^{-1} region - is diagnostic for bidentate versus monodentate DTC coordination and serves as a primary tool for complex characterization in Chapter 5 [33].

Studies by Venugopal et al. from Sri Krishnadevaraya University, Anantapur (India) demonstrated the synthesis and characterization of Cu(II) and Co(II) dithiocarbamate complexes through FTIR, UV-Visible, ^1H NMR, ESR, and TGA-DTA analysis, confirming bidentate S,S-coordination geometry and octahedral arrangement in these systems,

with the C–S stretching frequency appearing at 998 cm^{-1} in the free ligand shifting to 1025 cm^{-1} upon coordination [7].

3.5 Thiourea - Coordination Chemistry and Selectivity for Mercury

Thiourea $[(NH_2)_2C{=}S]$ is an organosulfur compound containing a thiocarbonyl group (C=S) flanked by two amino groups. It has been known to coordination chemists for decades and continues to attract considerable research attention owing to its structural versatility as a ligand, its accessibility, and its selective affinity for mercury and other soft metal ions [36].

3.5.1 Structure and Tautomerism

Thiourea exists in two tautomeric forms in equilibrium: the thione form [$(NH_2)_2C{=}S$], which is the dominant form in the solid state, and the thiol form [$NH_2(NH)C{-}SH$], which can become relevant in coordinating environments [36]. Both the sulfur atom and the two nitrogen atoms represent potential coordination sites, giving thiourea a formally ambidentate character; in practice, however, coordination through the sulfur lone pair dominates for soft metal ions, while N,S-bidentate chelation through both the sulfur and one nitrogen atom is observed for harder metal ions such as Cu(II), Ni(II), and Co(II) [36].

3.5.2 Coordination Modes

Three principal coordination modes have been identified for thiourea complexes [36]:

- **Monodentate S-coordination**: The sulfur lone pair donates to the metal ion exclusively, as observed for soft acid metal ions such as Hg^{2+}, Ag^{+}, and Au^{+}. In these complexes, the M–S bond length is short and the C=S bond is elongated relative to free thiourea, confirming electron donation from sulfur to the metal.
- **Bidentate S,N-coordination**: Both the sulfur atom and one nitrogen atom coordinate simultaneously to the metal centre, forming a four-membered chelate ring. This mode is observed for intermediate metal ions such as Cu^{2+} and Pd^{2+} and results in enhanced complex stability.
- **Bridging coordination**: Thiourea bridges two metal centres simultaneously through the sulfur atom, forming polynuclear complexes - a mode less relevant to aqueous analytical chemistry but important in materials science applications.

3.5.3 Selectivity of Thiourea for Mercury

Mercury(II) presents a unique coordination chemistry context in aqueous systems. As a borderline to soft Lewis acid, Hg^{2+} shows an overwhelming preference for sulfur donor ligands over nitrogen or oxygen donors - a selectivity rationalized through HSAB theory [31]. Thiourea exploits this preference effectively: the $[Hg(\text{thiourea})_4]^{2+}$ complex and related derivatives exhibit high thermodynamic stability, with the Hg–S bond strength arising from a combination of σ-donation from the sulfur lone pair and π-back-donation from mercury's filled d orbitals into sulfur's d orbitals [36].

This high stability and selectivity make thiourea analytically useful for the determination of mercury in wastewater through back-titration procedures, where excess thiourea is added to complex Hg^{2+} completely, and the residual thiourea is then titrated against a standardized solution [5]. In environmental samples from Indian industrial sites - particularly those associated with chloralkali plants, paint manufacturing, and pharmaceutical production - thiourea-based complexometric determination provides a cost-effective and accessible alternative to cold-vapour atomic absorption spectrometry (CV-AAS) for mercury quantification at concentrations approaching regulatory threshold levels [5].

CHAPTER 4

Mechanism of Complex Formation in Aqueous Medium

4.1 Introduction

The formation of metal–ligand complexes in aqueous medium is governed by a set of well-defined thermodynamic and kinetic principles that determine the rate, pathway, and extent of coordination bond formation between a metal ion and a chelating agent [36]. In environmental and analytical chemistry, understanding these mechanistic principles is essential for predicting how a chosen ligand will behave in the chemically complex matrix of industrial wastewater - where competing metal ions, variable pH, and diverse ionic species can all influence complex formation efficiency [33].

The present chapter examines the mechanism of complex formation in aqueous medium for four metal–ligand systems central to this investigation: the lead–dithiocarbamate system, the chromium–EDTA system, the mercury–thiourea system, and the copper–dithiocarbamate system. For each system, the reaction pathway, governing equilibria, pH dependence, and factors influencing stability are discussed in detail with reference to their analytical and remediation significance in the Indian wastewater context.

A general framework applicable to all four systems is the concept of ligand substitution, wherein water molecules from the primary hydration sphere of the aquated metal ion $[M(H_2O)_n]^{z+}$ are progressively displaced by the donor atoms of the incoming chelating ligand. The mechanism by which this substitution proceeds - whether associative (A), dissociative (D), or interchange (I_a or I_d) - depends on the electronic configuration and lability of the metal centre [36]. Labile metal ions such as Pb^{2+}, Cu^{2+}, and Hg^{2+}, whose water exchange rates exceed 10^7 s^{-1}, undergo rapid ligand substitution, forming complexes within seconds to minutes under favourable pH conditions. By contrast, inert metal ions such as Cr^{3+} exhibit water exchange rates of approximately 10^{-6} s^{-1}, making their complexation reactions kinetically slow - a distinction with profound analytical and treatment implications [36]

4.2 Lead–Dithiocarbamate Complex: Mechanism and Stability

4.2.1 Overview and Environmental Relevance

Lead (Pb^{2+}) is among the most pervasive heavy metal contaminants in Indian industrial wastewater, originating from electroplating operations, battery recycling plants, printing industry effluents, and lead-soldered pipeline corrosion in urban distribution networks [20]. The dithiocarbamate (DTC) ligand has been established as one of the most effective chelating agents for Pb^{2+} removal and analytical determination owing to the strong affinity between the soft sulfur donor atoms of DTC and the soft Lewis acid character of Pb^{2+}, which is rationaliz through Pearson's Hard and Soft Acids and Bases (HSAB) theory [33].

Importantly, dithiocarbamates form insoluble and distinctly coloured complexes with Pb^{2+} in water - a property exploited for both gravimetric removal and colorimetric determination of lead in wastewater samples [33].

4.2.2 Reaction Pathway and Mechanism

The formation of the lead–dithiocarbamate complex in aqueous medium proceeds through the following sequence of steps [33], [36]:

Step 1 - Synthesis of DTC ligand: The sodium dithiocarbamate is first prepared by the reaction of a secondary amine (R_2NH) with carbon disulfide (CS_2) under alkaline conditions:

$$R_2NH + CS_2 + NaOH \rightarrow R_2NCS_2Na + H_2O$$

Step 2 - Dissociation and speciation of Pb^{2+}: In aqueous solution, lead(II) exists predominantly as the hexaaqua ion $[Pb(H_2O)_6]^{2+}$ at acidic to near-neutral pH. At pH above 7, hydrolysis products $Pb(OH)^+$ and $Pb(OH)_2$ begin forming, reducing the concentration of free Pb^{2+} available for chelation [36]. Complexometric determination of Pb^{2+} is therefore optimal in the pH range 5–6, where the free ionic form predominates and DTC coordination is energetically favourable.

Step 3 - Ligand substitution: The dithiocarbamate anion ($R_2NCS_2^-$) approaches the Pb^{2+} centre through electrostatic attraction. Being a labile metal ion, Pb^{2+} rapidly undergoes water substitution through an interchange mechanism (I_d), where the first sulfur donor of DTC enters the coordination sphere as the first water molecule departs:

$$[Pb(H_2O)_6]^{2+} + R_2NCS_2^- \rightarrow [Pb(H_2O)_5(S_2CNR_2)]^+ + H_2O$$

Step 4 - Ring closure and chelate formation: The second sulfur atom of the same DTC molecule closes the chelate ring through a second intramolecular substitution, forming the three-membered Pb–S–C–S chelate ring:

$$[Pb(H_2O)_5(S_2CNR_2)]^+ \rightarrow [Pb(H_2O)_4(\kappa^2\text{-S,S-}S_2CNR_2)] + H_2O \text{ (ring closure)}$$

Step 5 - Formation of bis-complex: A second DTC molecule undergoes analogous coordination to give the neutral, sparingly soluble bis-complex:

$$[Pb(H_2O)_4(\kappa^2\text{-S,S-DTC})] + DTC^- \rightarrow [Pb(DTC)_2] + 4H_2O$$

The final product $[Pb(DTC)_2]$ is a neutral, four-coordinate complex of distorted square planar or pyramidal geometry, stabilized by the strong Pb–S covalent bonds and confirmed through FTIR spectroscopy. The characteristic ν(C–S) stretching frequency shifts from approximately 946 cm^{-1} in the free DTC ligand to 959–961 cm^{-1} in the Pb(II) complex, consistent with bidentate S,S-coordination and the reduction of C–S bond order upon coordination [41].

The ν(C=N) thiouride band in the range 1425–1468 cm^{-1} confirms partial double bond character of the C–N bond, which is retained on complexation [33].

4.2.3 Stability and pH Influence

The formation constant (log Kf) for Pb(II)–DTC complexes is in the range 17–21 depending on the nature of the N-substituents on the dithiocarbamate, reflecting thermodynamically stable complex formation [33]. The S,S-donor atom set and the soft-acid character of Pb^{2+} collectively drive a highly negative ΔG of formation. The stability of the complex decreases at strongly acidic pH (below 4) due to protonation of the DTC sulfur atoms, which reduces their donor ability and promotes decomposition of the dithiocarbamate ligand itself to CS_2 and the parent amine through acid hydrolysis [33]. An optimal pH window of 5.5–7.5 is therefore maintained for both the analytical determination and precipitation-based removal of Pb^{2+} using DTC ligands in Indian electroplating and battery industry effluents [20].

4.3 Chromium–EDTA Complex: Kinetics and pH Dependence

4.3.1 Overview and Environmental Relevance

Chromium(III) is the dominant form of chromium in tannery effluents across Indian leather processing clusters in Kanpur, Chennai, and Kolkata, where the chrome tanning process generates effluents containing Cr^{3+} at concentrations ranging from 1,000 to 4,000 mg/L before treatment - far exceeding the CPCB inland water discharge limit of 2.0 mg/L total chromium [24]. EDTA is widely employed as a chelating agent for the determination and speciation of Cr^{3+} in wastewater, and the $[Cr(EDTA)]^-$ complex has been extensively characterized. However, the Cr^{3+}–EDTA system is distinguished from other metal–EDTA systems by its exceptional kinetic inertness, arising from the d^3 electronic configuration of Cr^{3+}, which produces a high ligand field activation energy that substantially slows water exchange and hence complex formation [36].

4.3.2 Aqueous Speciation of Chromium(III)

In aqueous solution, Cr^{3+} exists as the hexaaqua ion $[Cr(H_2O)_6]^{3+}$, which undergoes progressive hydrolysis with increasing pH:

- $[Cr(H_2O)_6]^{3+}$ (pH < 4) - dominant aqua ion
- $[Cr(OH)(H_2O)_5]^{2+}$ - hydroxo mono-substituted species (pH 4–6)
- $[Cr(OH)_2(H_2O)_4]^{+}$ - dihydroxo species (pH 6–7)
- $Cr(OH)_3$ precipitate (pH > 7) - insoluble hydroxide

This speciation profile is critical for EDTA-based complexation, because only the free ionic and hydroxo species participate in ligand exchange reactions; the precipitation of $Cr(OH)_3$ at alkaline pH renders EDTA complexation negligible under those conditions and necessitates operation in the mildly acidic to neutral pH range (pH 3.8–6.5) for analytical determination [36].

4.3.3 Mechanism of Cr^{3+}–EDTA Complex Formation

The kinetics of Cr^{3+}–EDTA complex formation have been studied spectrophotometrically at 552 nm. The reaction is found to be first-order in Cr^{3+} and follows two concurrent mechanistic pathways - one pH-independent and one pH-dependent [36]:

Pathway 1 - Direct aqua ion substitution (pH-independent):

$[Cr(H_2O)_6]^{3+} + Y^{4-} \rightarrow [Cr(EDTA)]^- + 6H_2O$

This pathway involves stepwise dissociation of water molecules from the coordination sphere of $[Cr(H_2O)_6]^{3+}$ through a dissociative interchange (I_d) mechanism, with the rate-limiting step being the initial water dissociation that creates a coordinatively unsaturated intermediate:

$[Cr(H_2O)_6]^{3+}$ →^(slow) $[Cr(H_2O)_5]^{3+} + H_2O \rightarrow + Y^{4-} \rightarrow [Cr(EDTA)]^- + 5H_2O$

The rate constant k_1 for this pathway is approximately $4.4 \times 10^{-6}\ s^{-1}$, reflective of the extraordinarily slow water exchange rate of Cr^{3+} [36].

Pathway 2 - Hydroxo-assisted substitution (pH-dependent):

$[Cr(OH)(H_2O)_5]^{2+} + Y^{4-}$ →^(fast relative to Path 1) $[Cr(EDTA)]^- + 5H_2O + OH^-$

The monohydroxo species $[Cr(OH)(H_2O)_5]^{2+}$ undergoes water exchange significantly faster than the parent hexaaqua ion. This pathway shows an inverse first-order dependence on $[H^+]$ - meaning the rate increases with increasing pH - and first-order dependence on EDTA concentration.

The rate constant k_2 for hydroxo-assisted substitution is approximately $3.1 \times 10^{-4}\ s^{-1}$, roughly 100-fold greater than k_1 [36]. This explains the experimental observation that increasing pH from 3.3 to 4.7 substantially accelerates the Cr^{3+}–EDTA complexation rate, while higher ionic strength and elevated dielectric constant retard it through screening of ion-pair formation between the reactants [36].

The activation energy of the reaction has been determined as 37.1 ± 3 kJ mol^{-1}, and the large negative activation entropy ($\Delta S^\ddagger = -195.6 \pm 6$ J K^{-1} mol^{-1}) is indicative of a highly ordered transition state, consistent with an associative interchange mechanism in which the approaching EDTA occupies an outer-sphere coordination position before inner-sphere bond formation [36].

4.3.4 Product Stability and Environmental Significance

The $[Cr(EDTA)]^-$ complex, once formed, is thermodynamically very stable ($\log K_f = 23.4$) but undergoes an exceptionally slow ligand exchange reaction.

Studies on tannery wastewaters from Italian and Indian processing facilities have demonstrated that Cr(III)–EDTA complexes can persist in effluent streams for weeks without dissociation under ambient temperature and pH conditions [36].

This kinetic persistence means that Cr^{3+} present as $[Cr(EDTA)]^-$ in tannery effluents is not removed by conventional lime precipitation, since the EDTA chelation prevents precipitation of $Cr(OH)_3$ - a finding with direct regulatory implications for Indian tannery effluent treatment plants attempting to meet the CPCB discharge norm of 2.0 mg/L total Cr [24]

4.4 Mercury–Thiourea Complex: Formation and Selectivity

4.4.1 Overview and Environmental Relevance

Mercury(II) is one of the most toxic heavy metals encountered in Indian industrial wastewater, originating from chlor-alkali plants, thermometer manufacturing, dental amalgam processing, and pharmaceutical production units [23].

In the aqueous environment, Hg^{2+} undergoes microbially mediated methylation to form methylmercury - a persistent neurotoxin that bioaccumulates through aquatic food chains. The thiourea ligand, $(NH_2)_2C{=}S$, has been employed for the selective determination and complexometric titration of Hg^{2+} in wastewater for several decades owing to the remarkably high stability of Hg^{2+}–thiourea complexes relative to analogous complexes with nitrogen or oxygen donors [35].

4.4.2 Aqueous Speciation of Mercury(II)

In aqueous solution, Hg^{2+} exists predominantly as the diaqua ion $[Hg(H_2O)_2]^{2+}$ due to its preference for two-coordinate linear geometry, reflecting the filled $5d^{10}$ electronic configuration and strong relativistic effects that concentrate electron density in the 6s orbital, making Hg^{2+} one of the softest Lewis acids in the periodic table [35].

At alkaline pH, Hg^{2+} forms HgO precipitate or $[Hg(OH)]^+$ species, reducing the concentration available for thiourea coordination.

Thiourea complexation of Hg^{2+} is therefore most efficient under mildly acidic to near-neutral conditions (pH 3–6), within which the free diaqua ion predominates [35].

4.4.3 Mechanism of Hg^{2+}–Thiourea Complex Formation

Mercury(II), as an extremely labile metal ion with a water exchange rate exceeding $10^9\ s^{-1}$, undergoes essentially instantaneous ligand substitution. Mechanism of Hg^{2+}–thiourea complex formation involves rapid, stepwise displacement of coordinated water molecules by sulfur lone pair of successive thiourea molecules through associative interchange pathway [35]:

Step 1: First thiourea molecule coordinates through sulfur:

$[Hg(H_2O)_2]^{2+}$ + tu → $[Hg(tu)(H_2O)]^{2+}$ + H_2O (K_1)

Step 2: Second and subsequent thiourea molecules coordinate, with Hg^{2+} typically reaching a coordination number of four in the stable tetrathiourea complex:

$[Hg(tu)(H_2O)]^{2+}$ + tu → $[Hg(tu)_2]^{2+}$ + H_2O (K_2) $[Hg(tu)_2]^{2+}$ + 2tu → $[Hg(tu)_4]^{2+}$ (K_3, K_4)

The dominant product in excess thiourea is $[Hg(tu)_4]^{2+}$, a four-coordinate complex with linear or tetrahedral geometry depending on the concentration ratio of thiourea to Hg^{2+} [35]. The Hg–S bond in these complexes is characterized by a short bond length (Hg–S ≈ 2.36–2.41 Å), reflecting strong covalent sigma donation from the sulfur lone pair combined with π-back donation from Hg^{2+} 5d orbitals into thiourea's vacant d orbitals - a bonding interaction unique to soft acid–soft base pairs and responsible for the exceptional thermodynamic stability of Hg–thiourea complexes [35].

4.4.4 Selectivity of Thiourea for Hg^{2+} Over Competing Metal Ions

A critical analytical feature of the Hg^{2+}–thiourea system is its high selectivity for mercury in the presence of other heavy metal ions commonly found in industrial wastewater, including Cu^{2+}, Pb^{2+}, Zn^{2+}, and Cd^{2+} [35]. This selectivity arises from two complementary factors: First, the magnitude of the Hg^{2+}–thiourea stability constant is substantially greater than that for other metal ions. The overall formation constant β_4 for $[Hg(tu)_4]^{2+}$ is among the highest known for mercury with neutral sulfur donors, reflecting the unique soft-acid character of Hg^{2+} [35]. Second, the pH and concentration conditions under which Hg^{2+}–thiourea complexes are stable are generally unfavourable for competing divalent metal–thiourea complex formation, allowing selective determination of Hg^{2+} in the presence of Cu^{2+}, Pb^{2+}, and Mn^{2+} through appropriate pH control and use of auxiliary masking agents such as thioglycolic acid to suppress Cu^{2+} interference [5]. Thiourea-functionalized sorbents and porous aromatic frameworks (PAFs) modified with thiourea groups have demonstrated Hg^{2+} separation coefficients relative to competing ions as high as 7935 (Hg^{2+}/Zn^{2+}) and 3701 (Hg^{2+}/Cu^{2+}), quantitatively demonstrating the thermodynamic basis for mercury selectivity in aqueous medium [35].

4.5 Copper–Dithiocarbamate Complex: Reaction Pathway

4.5.1 Overview and Environmental Relevance

Copper(II) is among the most significant heavy metal contaminants discharged by electroplating industries, printed circuit board manufacturing units, textile dyeing factories, and agricultural run-off in India [23]. The dithiocarbamate ligand forms one of the most extensively studied and analytically well-characterized complexes with Cu^{2+}, known as the copper(II) bis-dithiocarbamate complex $[Cu(DTC)_2]$, which has been investigated for over 120 years across coordination chemistry, biological chemistry, materials science, and environmental engineering [65].

4.5.2 Electronic Configuration and Coordination Preference

Copper(II) possesses a d^9 electronic configuration, which renders it highly labile to ligand substitution (water exchange rate ~10^8 s^{-1}) and imparts a strong thermodynamic preference for square planar coordination geometry through Jahn-Teller distortion of the octahedral ligand field [65]. This d^9 configuration also makes Cu(II) dithiocarbamate complexes paramagnetic, with one unpaired electron readily detected by Electron Paramagnetic Resonance (EPR) spectroscopy - a valuable tool for confirming complex formation and determining the coordination geometry in solution [65].

4.5.3 Reaction Pathway

The formation of the Cu(II)–dithiocarbamate complex in aqueous medium proceeds as follows [33], [65]:

Step 1 - Speciation of Cu^{2+}: In aqueous solution at ambient pH, copper exists as $[Cu(H_2O)_6]^{2+}$, which undergoes facile Jahn-Teller distortion to give an effectively four-coordinate $[Cu(H_2O)_4]^{2+}$ species with two elongated axial water molecules [65].

Step 2 - Rapid monoligand complex formation: The DTC anion ($R_2NCS_2^-$) approaches the copper centre as a bidentate S,S-donor, displacing two equatorial water molecules in a rapid associative step:

$$[Cu(H_2O)_4]^{2+} + R_2NCS_2^- \rightarrow [Cu(H_2O)_2(\kappa^2\text{-S,S-DTC})]^+ + 2H_2O$$

In excess copper condition, monoligand complex $[Cu(DTC)]^+$ is dominant species [65].

Step 3 - Bis-complex formation: A second DTC anion coordinates to yield the stable, neutral bis-complex:

$$[Cu(H_2O)_2(DTC)]^+ + R_2NCS_2^- \rightarrow [Cu(DTC)_2] + 2H_2O$$

Step 4 - Geometry and solid-state structure: The product $[Cu(DTC)_2]$ is a neutral, mononuclear complex adopting a distorted square planar geometry in which all four sulfur atoms from two bidentate DTC ligands coordinate to the Cu^{2+} centre in a planar arrangement [65]. The Cu–S bond lengths are approximately 2.28–2.32 Å, consistent with strong covalent Cu–S bonds. In the solid state, monomeric and dimeric forms are known - the dimeric form being held together by intermolecular C···S interactions between adjacent complex molecules [65].

Step 5 - Redox lability (Cu^{2+}/Cu^+ equilibrium): A distinctive feature of the Cu–DTC system is its redox-active character. Cu(II) bis-dithiocarbamate complexes undergo a fully reversible one-electron oxidation to the d^8 Cu(III) complex $[Cu(DTC)_2]^+$, while reduction yields the Cu(I) species $[Cu(DTC)_2]^-$, which, as a d^{10} centre, undergoes structural rearrangement from square planar to tetrahedral [65]. This Cu^{2+}/Cu^+ redox lability is relevant in the environmental context because changes in the redox potential of wastewater - as encountered in alternating aerobic/anaerobic treatment zones - can shift the equilibrium of copper speciation and thereby affect the efficacy of DTC-based copper removal [65].

4.5.4 Stability, pH Sensitivity and Analytical Utility

The formation constant for $[Cu(DTC)_2]$ is in the range log Kf = 16–19, depending on the N-substituents. Under neutral to mildly acidic pH conditions (pH 5–8), the Cu–DTC complex is stable and precipitates as a brown-to-dark-green solid that can be gravimetrically determined or dissolved in organic solvents for spectrophotometric analysis at $\lambda\_max \approx 435$ nm [5], [65].

Under strongly acidic conditions (pH < 3), protonation of the DTC sulfur atoms leads to decomposition of the complex and release of Cu^{2+} back into solution - a property that is exploited for stripping and regeneration of DTC-modified adsorbents used in repeated-cycle wastewater treatment applications [33].

Studies employing zinc dimethyldithiocarbamate $[Zn(DMDC)_2]$ for the selective recovery of Cu(II) from strongly acidic wastewater demonstrated that copper displaces zinc from the dithiocarbamate framework through a transmetalation mechanism, owing to the stronger coordination bonds formed between Cu^{2+} and the DTC sulfur atoms compared to Zn^{2+}–DTC bonds. This approach achieved a Cu^{2+} removal efficiency of 99.6% under optimized conditions, with the recovered Cu–DTC precipitate further calcined at 800°C to yield high-purity CuO, demonstrating the potential for copper resource recovery from industrial effluents in addition to pollution control [65].

CHAPTER 5

Characterization of Metal Complexes

5.1 Introduction

The systematic characterization of metal–ligand complexes constitutes an indispensable phase in any study of coordination chemistry, bridging the gap between theoretical mechanistic understanding and experimental verification of complex structure, geometry, and stability [40]. Characterization serves multiple purposes in the present investigation: it confirms that complex formation has indeed occurred between the metal ion and the intended ligand, establishes the mode of coordination (monodentate vs. bidentate; the specific donor atoms involved), determines the geometry adopted by the central metal ion, and provides thermodynamic and physical data that predict the behaviour of the complex under environmental conditions encountered in Indian industrial wastewater [31].

For the four metal–ligand systems investigated in this study - lead–dithiocarbamate (Pb–DTC), chromium–EDTA (Cr–EDTA), mercury–thiourea (Hg–TU), and copper–dithiocarbamate (Cu–DTC) - a multi-technique characterization approach is employed, encompassing UV-Visible spectroscopy, Fourier Transform Infrared Spectroscopy (FTIR), Atomic Absorption Spectroscopy (AAS), thermogravimetric analysis (TGA/DTA), molar conductance measurements, and magnetic susceptibility studies [31], [41].Each technique interrogates a distinct property of the complex - its electronic transitions, molecular vibrations, elemental metal content, thermal stability, ionic character, and magnetic behaviour - and together they provide a comprehensive and internally consistent picture of the complex's structure.

5.2 UV-Visible Spectroscopy

5.2.1 Principle and Instrumentation

UV-Visible (UV-Vis) spectroscopy is among the most fundamental and widely applied techniques for the characterization of metal complexes in solution. The technique measures the absorption of ultraviolet (200–400 nm) and visible (400–800 nm) electromagnetic radiation by a sample dissolved in a suitable solvent, with the absorbed wavelengths corresponding to electronic transitions within the molecule [41]. For metal complexes, three categories of electronic transitions are analytically significant: d–d transitions, arising from the promotion of d-electrons between split d-orbital energy levels in the presence of a ligand field; ligand-to-metal charge transfer (LMCT) transitions, arising from the transfer of electron density from a filled ligand orbital to an empty metal d-orbital; and metal-to-ligand charge transfer (MLCT) transitions, the reverse process [41].

5.2.2 UV-Vis Characterization of the Four Complexes

Lead–Dithiocarbamate Complex [Pb(DTC)$_2$]: The UV-Vis spectrum of the Pb(II)–DTC complex in DMSO solution shows a strong absorption band in the range 290–320 nm, attributable to an intraligand π→π* transition of the dithiocarbamate moiety and an LMCT transition from the sulfur lone pair to the Pb^{2+} centre. Since Pb^{2+} is a d^{10} ion, no d–d transition is expected or observed. The yellow-to-orange colouration of the complex in solution is diagnostic and arises from the LMCT band tail extending into the visible region - a feature used for colorimetric detection of Pb^{2+} in water samples [41]. The intensity of this absorption band follows Beer–Lambert's law ($A = \varepsilon lc$) over the concentration range 0.5–10 mg/L, enabling construction of a calibration curve for Pb^{2+} quantification.

Chromium–EDTA Complex [Cr(EDTA)]$^-$: The d^3 electronic configuration of Cr^{3+} gives rise to three spin-allowed d–d transitions in an octahedral ligand field. The $[Cr(EDTA)]^-$ complex exhibits absorption bands at approximately 406 nm (ν_2: $^4A_2g \rightarrow {}^4T_1g$ transition) and 552 nm (ν_1: $^4A_2g \rightarrow {}^4T_2g$ transition), consistent with an octahedral coordination geometry around Cr^{3+} [36]. The precise positions of these bands shift relative to those of the hexaaqua ion $[Cr(H_2O)_6]^{3+}$ (absorption at 408 nm and 574 nm), confirming that ligand substitution has occurred and that EDTA exerts a stronger ligand field than water, consistent with its hexadentate donor set [36]. The kinetics of Cr^{3+}–EDTA complex formation can be conveniently monitored by tracking the progressive shift and development of the 552 nm absorption band over time using UV-Vis spectrophotometry, as employed in the kinetic studies.

Mercury–Thiourea Complex [Hg(TU)$_4$]$^{2+}$: Mercury(II) has a d^{10} electronic configuration and therefore exhibits no d–d transitions. The UV-Vis spectrum of $[Hg(tu)_4]^{2+}$ shows a strong absorption band near 220–240 nm arising from a π→π* intraligand transition of the thiourea C=S chromophore, with a significant red-shift (bathochromic shift) of approximately 15–20 nm relative to free thiourea - direct evidence of coordination through the sulfur atom, which reduces the C=S bond order and lowers the π* orbital energy [39]. This UV absorption band can be used to construct a calibration curve for Hg^{2+} determination by difference spectrophotometry - a method adapted for use with thiourea indicator in the back-titration procedure.

Copper–Dithiocarbamate Complex [Cu(DTC)$_2$]: Cu^{2+} carries a d^9 configuration and exhibits a broad, asymmetric d–d absorption band in the visible region at approximately 420–440 nm in the $[Cu(DTC)_2]$ complex, arising from the $^2Eg \rightarrow {}^2T_2g$ transition characteristic of a square planar geometry with Jahn-Teller distortion [40]. This strong absorption at ~435 nm imparts the characteristic dark brown-to-greenish-brown colouration of the complex and is the basis for its spectrophotometric determination in wastewater samples. The molar absorptivity (ε) of $[Cu(DTC)_2]$ at 435 nm is typically in the range 8,000–12,000 L mol^{-1} cm^{-1}, enabling quantification of Cu^{2+} at concentrations as low as 0.05 mg/L - well within the regulatory threshold of 3.0 mg/L (CPCB) [40].

5.3 Fourier Transform Infrared Spectroscopy (FTIR)

5.3.1 Principle and Spectral Regions

Fourier Transform Infrared Spectroscopy (FTIR) is based on the selective absorption of infrared radiation by molecular bonds undergoing characteristic vibrational motions - bond stretching, bending, rocking, wagging, and twisting - at frequencies determined by the masses of the bonded atoms and the force constant of the bond [41]. The mid-infrared region (4000–400 cm^{-1}) is most analytically relevant for metal complex characterization. The **functional group region** (4000–1500 cm^{-1}) contains diagnostic stretching frequencies of key bonds (N–H, C–H, C=O, C=N, C=S), while the **fingerprint region** (1500–400 cm^{-1}) contains complex overlapping bending vibrations unique to each molecule. Metal–ligand bond stretching vibrations (M–S, M–N, M–O) typically appear below 600 cm^{-1} and are particularly diagnostic for confirming coordination [41].

5.3.2 FTIR Characterization of the Four Complexes

Lead–Dithiocarbamate Complex: In the free DTC ligand, the characteristic ν(C–S) stretching frequency appears as a single sharp band at approximately 946 cm^{-1}. Upon coordination to Pb^{2+}, this band shifts to a higher frequency at 959–961 cm^{-1}, consistent with bidentate S,S-coordination of the DTC through both sulfur atoms, which strengthens the C–S bond through increased electron delocalization in the chelate ring [41]. The ν(C=N) thiouride band, present at 1468 cm^{-1} in the free ligand, shifts to 1429–1478 cm^{-1} in the complex, confirming partial double bond character of the C–N bond and S-coordination to Pb^{2+}. The absence of a band splitting pattern in the ν(C–S) region - specifically the appearance of a single rather than two bands - confirms symmetrical bidentate rather than monodentate coordination of the DTC anion [41]. The Pb–S stretching vibration appears as a new band below 500 cm^{-1} in the complex spectrum.

Chromium–EDTA Complex: The FTIR spectrum of the $[Cr(EDTA)]^-$ complex shows characteristic shifts in the carboxylate stretching vibrations of EDTA upon coordination. The asymmetric carboxylate stretch ν_as(COO^-) at approximately 1620 cm^{-1} in the free EDTA shifts to 1595–1610 cm^{-1} in the complex, while the symmetric carboxylate stretch ν_s(COO^-) at approximately 1390 cm^{-1} in free EDTA shifts to 1380–1400 cm^{-1} in the complex. The separation Δν = ν_as – ν_s decreases from ~230 cm^{-1} in free EDTA to ~200 cm^{-1} in the complex, consistent with bidentate coordination of the carboxylate groups. New bands appearing at 590–620 cm^{-1} and 450–480 cm^{-1} are attributed to ν(Cr–O) and ν(Cr–N) stretching vibrations respectively, confirming metal–ligand bond formation [36]. The broad O–H stretching band from coordinated or lattice water molecules, if present, appears at 3300–3500 cm^{-1}.

Mercury–Thiourea Complex: The most diagnostic FTIR change upon thiourea coordination to Hg^{2+} is the significant elongation of the C=S bond, reflected in a decrease of the ν(C=S) stretching frequency from approximately 730 cm^{-1} in free thiourea to 650–680 cm^{-1} in the Hg–thiourea complex [39]. This downward shift confirms that the sulfur lone pair has been donated to Hg^{2+}, weakening the C=S bond by reducing electron density at sulfur. Simultaneously, the N–H stretching bands at 3100–3350 cm^{-1} and the N–C–N bending vibration near 1085 cm^{-1} in free thiourea shift slightly upon coordination, indicating a redistribution of electron density from the C=S bond toward the C–N bond through the thioureide resonance mechanism. The appearance of new bands in the 300–350 cm^{-1} region is attributed to ν(Hg–S) stretching vibrations, confirming Hg–S bond formation [39].

Copper–Dithiocarbamate Complex: The FTIR spectrum of $[Cu(DTC)_2]$ displays characteristic thiouride band ν(C=N) in the range 1425–1456 cm^{-1}, which shifts to higher frequency (1446–1499 cm^{-1}) in the complex - confirming the partial double bond character of the C–N bond and bidentate S,S-coordination [40]. The ν(C–S) stretching vibrations appear as a single strong band in the range 1012–1043 cm^{-1} in the complex, consistent with symmetrical bidentate coordination (a single band confirms both C–S bonds are equivalent, i.e., bidentate mode). The presence of the ν(Cu–S) stretching vibration below 500 cm^{-1} further confirms Cu–S bond formation. Studies at Banaras Hindu University (BHU), Varanasi characterized analogous transition metal DTC complexes and reported molar conductance values of 10–20 ohm^{-1} cm^3 mol^{-1} in DMSO, confirming non-electrolytic character of the neutral bis-DTC complexes [42].

5.4 Atomic Absorption Spectroscopy (AAS)

5.4.1 Principle and Instrumentation

Atomic Absorption Spectroscopy (AAS) is the most widely employed instrumental technique for the quantitative determination of metal ions in environmental, industrial, and biological samples [43]. The technique is based on the absorption of characteristic radiation by free ground-state atoms of the analyte element in the atomized sample. Each metal element absorbs radiation at a unique wavelength corresponding to its specific ground state electronic transition - for example, Pb at 217 nm, Cr at 357.9 nm, Hg at 253.7 nm, and Cu at 324.7 nm [43]. The degree of absorption, measured as absorbance A, is directly proportional to the concentration of the element in the sample (Beer-Lambert's Law), enabling quantification through comparison with a standard calibration curve.

Two principal modes of AAS are employed in heavy metal analysis of wastewater: Flame AAS (FAAS), which uses an air-acetylene or nitrous oxide-acetylene flame to atomize the sample and is suitable for metal concentrations in the range 0.1–100 mg/L; and Graphite Furnace AAS (GFAAS), which employs a resistively heated graphite tube for atomization, offering detection limits 100–1000-fold lower than FAAS (typically 0.001–0.01 µg/L) and hence better suited for trace metal determination at concentrations near or below regulatory thresholds [43].

5.4.2 AAS as Validation Tool for Complexometric Determination

In the present investigation, AAS serves a dual role: as a primary tool for the quantitative determination of metal concentrations in wastewater samples , and as a validation method for the elemental analysis data obtained from the synthesized metal complexes. For elemental validation of the complexes, the metal content determined by AAS after acid digestion of the complex is compared against the theoretical metal content calculated from the proposed molecular formula. Agreement within ±0.5% is considered acceptable confirmation of the proposed structure.

A study conducted at Bisalpur Dam, Rajasthan, using AAS for seasonal monitoring of heavy metals in drinking water demonstrated distinct site-wise and seasonal variations in Pb, Cu, Ni, Fe, Cr, Cd, and Zn concentrations over 2021–23, with all metals within BIS and WHO limits at that site [43]. However, analogous studies at Indian industrial sites - particularly in electroplating clusters of Ludhiana (Punjab) and in tannery zones of Kanpur - have reported Cr and Pb concentrations well above permissible limits by AAS, corroborating the complexometric data discussed in Chapter 6 [11].

5.5 Molar Conductance Measurements

5.5.1 Principle and Interpretation

Molar conductance (Λ_m) measurements - made by dissolving the metal complex at a known concentration (typically 10^{-3} mol/L) in a suitable solvent such as DMSO or DMF and measuring the electrical conductivity - provide direct information about the ionic character of the complex [42]. Electrolytic complexes, which dissociate into cations and anions in solution, exhibit high molar conductance values, while non-electrolytic neutral complexes show low values. The interpretation follows the empirical ranges established by Geary (1971):

- Non-electrolyte: Λ_m = 10–20 ohm^{-1} cm^2 mol^{-1} (neutral complex, no dissociation)
- 1:1 electrolyte: Λ_m = 70–100 ohm^{-1} cm^2 mol^{-1} (one ionic pair formed)
- 1:2 electrolyte: Λ_m = 160–220 ohm^{-1} cm^2 mol^{-1} (two ionic pairs formed)

5.5.2 Results for the Four Complexes

For the neutral bis-complexes $[Pb(DTC)_2]$ and $[Cu(DTC)_2]$, which carry no net ionic charge, molar conductance values in the range 10–20 ohm^{-1} cm^2 mol^{-1} in DMSO confirm their non-electrolytic character - consistent with the proposed neutral, four-coordinate structures [42]. The $[Cr(EDTA)]^-$ complex, which bears a net charge of –1, exhibits low conductance consistent with a 1:1 electrolyte when its sodium or potassium salt ([Na][Cr(EDTA)]) is dissolved, whereas the neutral Hg^{2+} complexes with thiourea ($[Hg(tu)_2]Cl_2$) show higher conductance due to chloride counter-ion dissociation.

These molar conductance data provide independent corroboration of the coordination mode and ionic character conclusions drawn from FTIR and UV-Vis spectroscopy [42].

5.6 Magnetic Susceptibility Measurements

5.6.1 Principle

Magnetic susceptibility (χ_m) measurement determines whether a metal complex is paramagnetic (possessing unpaired electrons, attracted to a magnetic field) or diamagnetic (all electrons paired, repelled by a magnetic field) [44]. For transition metal complexes, the effective magnetic moment (μ_{eff}) is calculated from the measured molar magnetic susceptibility using the spin-only formula:

$$\mu_{eff} = \sqrt{n(n+2)} \text{ B.M.}$$

where n is the number of unpaired electrons and B.M. represents Bohr Magnetons. The experimentally measured μ_{eff} is then compared against the spin-only value to establish the number of unpaired electrons and confirm the proposed electronic configuration and geometry of the complex [44].

5.6.2 Results and Interpretation

The magnetic susceptibility data for the four complexes are directly informative about their electronic structure:

Pb^{2+} (d^{10}): Pb^{2+} is diamagnetic with no unpaired d-electrons. The $[Pb(DTC)_2]$ complex is expected to be diamagnetic ($\mu_{eff} \approx 0$ B.M.), consistent with a d^{10} configuration. This is confirmed experimentally by the absence of any attraction in the Gouy balance measurement [44].

Cr^{3+} (d^3): Three unpaired electrons in a d^3 configuration give a spin-only magnetic moment of $\mu_{eff} = \sqrt{(3\times5)} = 3.87$ B.M. Experimentally measured μ_{eff} values for $[Cr(EDTA)]^-$ in the range 3.70–3.90 B.M. confirm the high-spin d^3 configuration and are fully consistent with an octahedral geometry [44].

Hg^{2+} (d^{10}): Mercury(II) is diamagnetic ($\mu_{eff} \approx 0$ B.M.), and all Hg–thiourea complexes are expected to be diamagnetic - confirmed by the negative Gouy balance response [44].

Cu^{2+} (d^9): One unpaired electron in a d^9 configuration gives spin-only $\mu_{eff} = \sqrt{(1\times3)} = 1.73$ B.M. Experimentally measured values for $[Cu(DTC)_2]$ in the range 1.70–1.90 B.M. confirm one unpaired electron and are consistent with the proposed square planar or distorted square planar geometry. The EPR spectrum of $Cu(DTC)_2$ shows a characteristic signal at $g \approx 2.06$, further confirming the Cu^{2+} (d^9) paramagnetic centre [40].

A study from Banaras Hindu University (BHU), Varanasi characterized Cu(II), Ni(II), and Zn(II) complexes of Schiff base and DTC-type ligands using a comprehensive suite of techniques including IR, UV-Vis, TGA, magnetic susceptibility, and molar conductance, obtaining μ_{eff} values of 1.75–1.85 B.M. for Cu(II) complexes - in close agreement with the values observed in the present work and consistent with square planar geometry [42].

5.7 Thermogravimetric Analysis (TGA/DTA)

5.7.1 Principle

Thermogravimetric Analysis (TGA) measures the change in mass of a sample as a function of temperature, while Differential Thermal Analysis (DTA) simultaneously records the heat flow associated with endothermic and exothermic events [41]. Together, TGA/DTA reveal the thermal decomposition sequence of a metal complex - including the temperatures of dehydration, ligand loss, and ultimate decomposition to the metal oxide or sulfide residue - and provide information about the thermal stability and structural composition of the complex.

5.7.2 TGA/DTA Results for the Four Complexes

For the dithiocarbamate complexes ([Pb(DTC)$_2$] and [Cu(DTC)$_2$]), the TGA curves characteristically show three to four decomposition stages between 40–800°C:
Stage 1 (Room temperature to ~200°C): No mass loss, confirming that the complexes contain no coordinated or lattice water molecules - consistent with the absence of O–H stretching bands in the 3300–3500 cm^{-1} region of the FTIR spectrum [41].
Stage 2 (~200–350°C): Loss of the first DTC ligand as CS_2 and the parent amine fragment, accompanied by an exothermic DTA peak - indicating thermally driven oxidative decomposition of the DTC chelate ring.
Stage 3 (~350–600°C): Loss of the second DTC ligand, with further mass reduction.
Stage 4 (>600°C): Residue stabilizes as the metal sulfide (PbS or CuS) or oxide, confirmed by powder X-ray diffraction. Copper–DTC complexes are well-documented as single-source precursors (SSPs) for CuS nanoparticles, decomposing to hexagonal covellite (CuS) phase at temperatures above 350°C [40].
For the [Cr(EDTA)]$^-$ complex, TGA shows an initial mass loss at 100–150°C corresponding to dehydration of coordinated water molecules (if present as the monohydrate), followed by progressive decomposition of the EDTA ligand at 250–450°C, and final conversion to Cr_2O_3 above 700°C. The relatively high decomposition temperature of the [Cr(EDTA)]$^-$ complex, compared to simpler Cr^{3+} salts, reflects the thermodynamic stability conferred by the hexadentate EDTA chelation [36].

CHAPTER 6

Application of Complexometric Approach in Wastewater Treatment

6.1 Introduction

The application of the complexometric approach in wastewater treatment represents the practical culmination of the theoretical, mechanistic, and characterization work presented in the preceding chapters. Complexometric methods serve a dual function in environmental chemistry: as an analytical tool for the precise quantitative determination of heavy metal ions in wastewater samples, and as a treatment strategy through which chelating agents are employed to selectively capture and remove dissolved metal ions from industrial effluents [46]. This dual utility - achieved with relatively simple laboratory equipment, cost-accessible reagents, and analytically well-established protocols - renders complexometric methods particularly suited to the resource and infrastructure constraints prevalent across Indian environmental monitoring and industrial effluent treatment settings [5].

India's industrial sector generates a diverse and chemically complex range of wastewater streams - from chromium-laden tannery effluents in Kanpur and Chennai, to copper- and lead-rich electroplating discharges in Ludhiana and Aurangabad, to mercury-containing pharmaceutical and chloralkali plant effluents in Vadodara and Bhopal [24].

This chapter details the step-by-step application of complexometric titration for the estimation of key heavy metals (Pb^{2+}, Mn^{2+}, Hg^{2+}, Cu^{2+}), the determination of water hardness (Ca^{2+} and Mg^{2+}) in Indian water samples, the application of complexometric principles in industrial effluent treatment with case studies from Indian industrial clusters, and a critical comparison of the complexometric approach against conventional analytical and treatment methods.

6.2 Complexometric Titration for Heavy Metal Estimation - Procedure and Protocol

6.2.1 General Principle of Complexometric Titration

Complexometric titration involves the controlled addition of a standardized solution of a chelating agent - predominantly EDTA (as its disodium salt, Na_2H_2Y) - to a sample solution containing the metal ion(s) to be determined [46]. EDTA forms stable, soluble, 1:1 complexes with virtually all divalent and trivalent metal ions of environmental significance. The endpoint of the titration is detected using a metallochromic indicator - a chelating dye that forms a weakly coloured complex with the metal ion at the start of the titration, which is then displaced by the more stable EDTA complex at the equivalence point, producing a sharp colour change [32]. The volume of EDTA consumed at the endpoint is used to calculate the metal ion concentration in the original sample [46].

6.2.2 Estimation of Lead (Pb^{2+}) by EDTA Complexometric Titration

The determination of Pb^{2+} in industrial wastewater samples using EDTA titration follows a well-established protocol documented for Indian industrial effluent monitoring [5]:
Reagents: 0.01 M standardized EDTA solution; pH 5.5 acetate buffer (sodium acetate + acetic acid); Xylenol Orange (XO) indicator, 0.5% aqueous solution; potassium cyanide (KCN) masking agent (to mask Cu^{2+}, Zn^{2+}, Ni^{2+} if present).

Procedure: A 50 mL aliquot of the filtered, acid-digested wastewater sample is taken in a 250 mL conical flask. The pH is adjusted to 5.5 by adding acetate buffer. KCN masking solution (1 mL) is added to suppress interference from Cu^{2+} and Zn^{2+}. Three to four drops of Xylenol Orange indicator are added - the solution turns red, indicating formation of the Pb^{2+}–XO complex. EDTA is added from a burette with constant swirling until the colour changes sharply from red to yellow - the endpoint indicating complete chelation of all Pb^{2+} by EDTA, displacing the indicator [5].

6.2.3 Estimation of Manganese (Mn^{2+}) by EDTA Complexometric Titration

Manganese determination by EDTA titration employs PAN (1-(2-pyridylazo)-2-naphthol) as the indicator in a buffered medium at pH 10 (ammonia–ammonium chloride buffer) [46]. Mn^{2+} forms a red Mn–PAN complex at the start; addition of EDTA progressively chelates Mn^{2+} until the endpoint is reached, at which point the red colour shifts to yellow (free PAN). The method has been validated for steel plant effluents from Indian industrial zones, where Mn^{2+} concentrations frequently arise from steel production, battery manufacturing, and chemical sector discharges [46].

Interference from Ca^{2+} and Mg^{2+} - both titratable with EDTA at pH 10 - is addressed by performing a separate calcium hardness titration and subtracting its contribution, or by using a selective masking strategy.

6.2.4 Estimation of Mercury (Hg^{2+}) by Back-Titration with Thiourea

Direct EDTA titration of Hg^{2+} is problematic because the $[Hg(EDTA)]^{2-}$ complex, while thermodynamically stable, undergoes relatively slow equilibration at room temperature due to Hg^{2+}'s tendency to form competing polyhydroxo species [5]. A back-titration procedure using thiourea is therefore employed:
A known excess of standardized thiourea solution is added to the sample, ensuring complete complexation of all Hg^{2+} as $[Hg(tu)_4]^{2+}$. The residual (unreacted) thiourea is then titrated against a standard $Hg(NO_3)_2$ solution, using Diphenylcarbazone as indicator. The Hg^{2+} concentration in the original sample is calculated by difference [5].
This back-titration approach achieves a detection limit of approximately 0.01–0.05 mg/L for Hg^{2+} - approaching the BIS IS:10500 limit of 0.001 mg/L for mercury in drinking water. For samples near or below this threshold, the complexometric approach is supplemented by Cold Vapour AAS as described in Chapter 5 [5].

6.2.5 Estimation of Copper (Cu^{2+}) by EDTA Titration

Cu^{2+} is determined by direct EDTA titration at pH 4–5 using PAN or Murexide (ammonium purpurate) as indicator, within an acetate or citrate buffer system [46]. The Cu^{2+}–PAN complex imparts a deep violet colouration at the start of the titration; addition of EDTA displaces PAN from the weaker Cu–PAN complex to form the more stable Cu–EDTA complex, and the solution turns yellow at the endpoint. An alternative approach involves back-titration using excess EDTA followed by back-titration of the surplus with standardized zinc or lead solution. This method is preferred when the sample matrix contains competing metal ions at high concentrations, as is common in electroplating effluents from Indian industrial estates in Faridabad, Ludhiana, and Pune [46].

6.3 Determination of Water Hardness in Indian Water Samples

6.3.1 Significance of Hardness Determination

Water hardness - primarily due to dissolved Ca^{2+} and Mg^{2+} ions - is one of the most routinely determined parameters in water quality monitoring across India. Hard water impairs the effectiveness of soaps and detergents, causes scale deposition in boilers and pipeline infrastructure, reduces heat transfer efficiency in industrial cooling systems, and contributes to urinary tract complications with chronic consumption [47]. BIS IS:10500:2012 specifies an acceptable total hardness limit of 300 mg/L as $CaCO_3$ and a permissible limit of 600 mg/L in the absence of an alternate source.

A 2023 study published in the International Journal of Novel Research and Development (IJNRD) - an Indian indexed journal - documented the determination and removal of hardness from water samples across multiple Indian states using the EDTA complexometric method, confirming its reliability and reproducibility for routine monitoring [47].

6.3.2 EDTA Complexometric Procedure for Total Hardness

Reagents: 0.01 M standardized EDTA (disodium salt); pH 10 ammonia–ammonium chloride buffer; Eriochrome Black T (EBT) indicator (dissolved in ethanol or as a tablet); standardized 0.01 M $CaCl_2$ solution for EDTA standardization.

Procedure: A 50 mL water sample is pipetted into a conical flask. The pH is adjusted to 10.0 ± 0.1 using the ammonia buffer (3 mL). Three to four drops of EBT indicator are added - the solution turns wine red, indicating formation of the Ca^{2+}/Mg^{2+}–EBT complex. The burette is filled with standardized EDTA and titrated dropwise with swirling until the wine red colour changes sharply to steel blue - the endpoint [47].

The volume of EDTA consumed (V_1) corresponds to total hardness ($Ca^{2+} + Mg^{2+}$).

Calcium hardness: A separate 50 mL sample is titrated at pH 12–13 (achieved by adding 2 mL of 8 M NaOH), at which Mg^{2+} precipitates as $Mg(OH)_2$ and does not react with EDTA.

Murexide indicator is used; the endpoint is a colour change from pink to purple. The volume V_2 gives calcium hardness directly [47].
Magnesium hardness = Total hardness − Calcium hardness (calculated by difference).
Expression: Hardness as mg/L $CaCO_3$ = (V_EDTA × M_EDTA × 100,050 mg/mol $CaCO_3$ × 1000) / V_sample (mL).

6.4 Indian Industrial Case Studies - Application of Complexometric Monitoring

6.4.1 Tannery Effluent - Kanpur and Chennai Clusters

The tannery industry represents India's single most significant source of chromium contamination in industrial wastewater. India is the second largest producer of footwear and leather garments globally, with Kanpur (Uttar Pradesh) and Chennai (Tamil Nadu) being the primary leather processing centres, together processing 700,000 tonnes of hides and skins annually to produce two billion square feet of leather [48]. The chrome tanning process generates 2–3 kL of chrome-bearing effluent per tonne of hide processed, with raw Cr^{3+} concentrations of 1,000–4,000 mg/L at the tanning stage - far exceeding the CPCB discharge norm of 2 mg/L [48].

A pivotal Indian study by Das et al. (2023), conducted by researchers at CSIR-Central Leather Research Institute, Chennai, and its Kanpur regional centre, visited approximately 400 tanneries and 21 centralized effluent treatment plants (CETPs) across India over several years, documenting waste management practices and effluent characteristics [48]. The study found that while most large tanneries had functional CETPs employing lime precipitation for Cr removal, a significant proportion of small-cluster tanneries - particularly in the Jajmau area of Kanpur - discharged partially treated or untreated effluent into open drains feeding into the Ganga river system [48].

Chromium concentrations in Jajmau drain were documented at levels 10–15-fold above permissible limits, with EDTA-based complexometric back-titration serving as the primary method for routine Cr monitoring by State Pollution Control Board (SPCB-UP) lab [37].
A bioremediation study of tannery wastewaters collected from Jajmau, Kanpur and Pallavaram CETP, Chennai, demonstrated that indigenous bacterial strains (notably Citrobacter freundii) achieved chromium reduction of 73%, BOD reduction of 86%, and COD reduction of 80% - confirming that combined biological and chemical treatment can achieve compliance with CPCB norms [48]. Complexometric determination of residual Cr at each treatment stage enabled precise monitoring of treatment efficiency, with EDTA back-titration confirming that post-biological treatment Cr concentrations fell below 1.5 mg/L at the Chennai CETP - within the permissible discharge norm [48].

6.4.2 Pilot-Scale Chromium Recovery from Kanpur Tannery Wastewater

A pilot-scale chromium recovery study by a team from Jajmau, Kanpur, employing chemical precipitation with lime [$Ca(OH)_2$] and MgO at doses of 1,500–2,500 mg/L, documented Cr removal efficiencies exceeding 99% from raw tannery wastewater at initial Cr concentrations of approximately 5,010 mg/L [49]. The recovered chromium hydroxide cake was subsequently recycled as basic chromium sulphate for re-tanning - demonstrating circular economy potential in the Indian tannery sector. Throughout this study, EDTA complexometric titration served as the routine analytical method for total Cr monitoring at each precipitation stage, with results validated against ICP-OES for key samples. The complexometric method consistently yielded Cr values within 5% of ICP-OES measurements - confirming its adequacy for operational monitoring in Indian industrial settings [49].

6.4.3 Mahanadi River Basin - Water Quality Monitoring by Complexometric Methods

A spatiotemporal water quality study conducted across 16 sampling locations along the Mahanadi River basin, Odisha, over the period 2020–2024, employed EDTA complexometric titration as the primary method for water hardness and divalent metal ion monitoring, supplemented by UV-Vis spectrophotometry for heavy metal quantification [50]. The study collected water samples during monsoon periods from sites spanning agricultural, industrial, and municipal zones. Results indicated that over 62.5% of examined samples exceeded the allowable threshold for total nitrogen, and that locations near industrial zones showed significantly elevated divalent metal concentrations consistent with industrial effluent ingress [50]. The EDTA hardness method provided reliable, reproducible results across all 16 sites, reinforcing its suitability as a field-adaptable analytical technique for the National Water Quality Monitoring Programme (NWQMP) of CPCB.

6.4.4 Electroplating Industry - Lead and Copper Monitoring

Electroplating industries - concentrated in industrial estates across Ludhiana (Punjab), Faridabad (Haryana), Aurangabad (Maharashtra), and Chennai (Tamil Nadu) - discharge effluents containing Pb^{2+}, Cu^{2+}, Ni^{2+}, and Cr^{6+} at concentrations that frequently exceed CPCB norms without adequate treatment [20]. A complexometric study on identification and estimation of Pb^{2+}, Mn^{2+}, and Hg^{2+} in wastewater samples collected from various Indian industrial discharge sites applied EDTA titration with Xylenol Orange (Pb), PAN (Mn), and back-titration with thiourea (Hg) [46]. The study documented Pb^{2+} at 0.8–4.5 mg/L (CPCB norm: 0.1 mg/L), Mn^{2+} at 1.2–3.8 mg/L (CPCB norm: 2.0 mg/L), and Hg^{2+} at 0.005–0.018 mg/L (CPCB norm: 0.01 mg/L) at various industrial effluent sampling sites [46]. These data confirm widespread non-compliance with CPCB discharge norms in the sampled industrial clusters, underscoring the public health and environmental significance of robust analytical monitoring programmes using accessible complexometric methods. Similar findings have been reported along the Visakhapatnam coastal industrial belt, where AAS analysis documented elevated concentrations of lead, chromium, and copper in environmental samples, further validating the necessity of routine heavy metal surveillance in Indian industrial zones [45].

6.5 Complexometric Treatment Strategies - EDTA-Enhanced Metal Removal

Beyond analytical determination, complexometric principles have been extended to the design of wastewater treatment systems that exploit the high metal–ligand affinity of chelating agents for selective metal capture and removal.

6.5.1 DTC-Functionalized Adsorbents

Dithiocarbamate-functionalized adsorbents - prepared by grafting DTC groups onto activated carbon, silica, cellulose, or chitosan surfaces - have demonstrated high efficiency for the removal of Pb^{2+}, Cu^{2+}, Cd^{2+}, and Hg^{2+} from synthetic and real industrial wastewater samples [38]. The metal removal mechanism is analogous to the complexation reactions described in Chapter 4: the immobilized DTC sulfur donor atoms coordinate to the target metal ion in bidentate S,S-fashion, forming a stable surface complex that immobilizes the metal on the adsorbent surface and removes it from solution. A study employing dimethyldithiocarbamate-modified adsorbent for selective Cu^{2+} recovery from strongly acidic electroplating wastewater achieved Cu^{2+} removal efficiency of 99.6% through a metal displacement mechanism, with the recovered Cu–DTC precipitate subsequently calcined to yield high-purity CuO - demonstrating both pollution control and resource recovery [40].

6.5.2 EDTA-Enhanced Adsorption and Soil Remediation

EDTA has been employed not only as a titrant but also as a chelation-mobilization agent in soil remediation applications adjacent to Indian tannery zones. Studies from CPCB-monitored sites around Kanpur have documented the use of EDTA and citric acid in combination to mobilize Cr^{3+} from contaminated soils for subsequent removal by adsorption or precipitation, achieving soil Cr depletion of 45–65% over three application cycles [25]. While this application differs from in-solution complexometric treatment, it illustrates the versatility of the chelation principle across environmental remediation contexts.

6.5.3 EDTA–Membrane Coupled Systems

Emerging treatment strategies combine EDTA chelation with membrane separation (nanofiltration or reverse osmosis) to achieve selective and regenerable heavy metal removal from complex industrial effluent matrices. In these systems, EDTA is added to the effluent to form soluble metal–EDTA complexes, which are then rejected by the nanofiltration membrane while competing anions and small molecules pass through. The EDTA–metal complex is then acidified to release the metal ion for recovery, and the regenerated EDTA is recycled [17]. Such combined systems are being evaluated for deployment at Indian industrial effluent treatment plants, particularly in the textile and electroplating sectors, under National Water Mission

Chapter 7: General Conclusion and Future Scope

7.1 Summary of Findings Across All Chapters

The present investigation systematically evaluated the complexometric approach - employing ethylenediaminetetraacetic acid (EDTA), dithiocarbamate (DTC), and thiourea - as an analytical and treatment strategy for heavy metal contamination in wastewater, with exclusive reference to the Indian industrial context.

Chapter 1 established that India faces a severe water pollution crisis: approximately 72% of urban wastewater is discharged untreated, and 37 river stretches remain critically polluted with heavy metals, notably the Kanpur–Varanasi stretch of the Ganga River due to tannery effluents [2][3][8]. Conventional treatment methods fail to selectively remove trace heavy metals, justifying complexometric alternatives [4][7][15].

Chapter 2 examined the aqueous chemistry of Pb^{2+}, Cr^{3+}/Cr^{6+}, Hg^{2+}, Cu^{2+}, and Mn^{2+}, including their industrial sources (tanneries, electroplating, chloralkali plants, textile dyeing) and health impacts. The Sukinda Valley (Cr^{6+} exceeding standards by >2×) and East Kolkata Wetlands (mercury bioaccumulation in fish) exemplify contamination severity [22][28].

Chapter 3 provided the theoretical foundation of ligand chemistry. EDTA is a hexadentate chelator (log Kf: 13.9 for Mn^{2+} to 23.4 for Cr^{3+}); DTC is an S,S-bidentate soft base selective for Pb^{2+}, Hg^{2+}, and Cu^{2+} (HSAB principle); thiourea is highly selective for Hg^{2+} [31][33][34][36].

Chapter 4 elucidated mechanistic pathways. Pb–DTC forms a neutral $[Pb(DTC)_2]$ bis-complex (optimal pH 5.5–7.5) [33][41]. Cr–EDTA exhibits exceptional kinetic inertness (water exchange $\approx 10^{-6}\ s^{-1}$), with the hydroxo-assisted pathway ($k_2 \approx 3.1 \times 10^{-4}\ s^{-1}$) being 100× faster than direct aqua ion route ($k_1 \approx 4.4 \times 10^{-6}\ s^{-1}$) [36]. Hg–thiourea forms stable $[Hg(tu)_4]^{2+}$ with high selectivity [35][39]. Cu–DTC yields square planar $[Cu(DTC)_2]$ (μ_{eff} = 1.70–1.90 B.M.; λ_{max} = 435 nm; ε = 8,000–12,000 L mol^{-1} cm^{-1}) [40][42].

Chapter 5 confirmed structures via characterization. FTIR showed ν(C–S) shift from 946 cm^{-1} (free DTC) to 959–961 cm^{-1} (Pb–DTC), confirming bidentate S,S-coordination [41]. Molar conductance (10–20 ohm^{-1} cm^2 mol^{-1}) confirmed non-electrolytic character [42]. TGA indicated thermal stability up to 200°C [40].

Chapter 6 applied these findings. EDTA titration protocols were established for Pb^{2+} (Xylenol Orange, pH 5.5), Mn^{2+} (PAN, pH 10), Cu^{2+} (PAN, pH 4–5), and Hg^{2+} (thiourea back-titration) [5][46]. Water hardness determination (EBT, pH 10) was validated [47]. Case studies from Kanpur and Chennai tannery clusters confirmed complexometric titration as the primary analytical method for Cr monitoring by SPCBs; raw Cr^{3+} (1,000–4,000 mg/L) was reduced to <2.0 mg/L after treatment [24][48][49]. The Jajmau pilot study showed >99% Cr removal, with complexometric results within 5% of ICP-OES [49]. The Mahanadi River basin study (16 sites, 2020–2024) validated the EDTA hardness method for the National Water Quality Monitoring

Programme [50]. Electroplating monitoring documented Pb^{2+} at 0.8–4.5 mg/L (norm: 0.1 mg/L) and Hg^{2+} at 0.005–0.018 mg/L (norm: 0.01 mg/L), confirming widespread non-compliance [46].

7.2 Effectiveness of Complexometric Methods

Analytical Effectiveness: EDTA titration provides precise, reproducible, and cost-effective quantification of Pb^{2+}, Mn^{2+}, Cu^{2+}, and total hardness, with detection limits of 0.5–1.0 mg/L (direct) and 0.01–0.05 mg/L for Hg^{2+} (back-titration) [5][46]. The technique requires only basic glassware and reagents, making it suitable for resource-limited Indian laboratories. Results agree with AAS/ICP-OES within 5–10%, confirming adequacy for regulatory monitoring [43][49].

Treatment Effectiveness: DTC-functionalized adsorbents achieved >95% removal of Pb^{2+} and Cu^{2+} at pH 5.5–7.5, with metal–DTC precipitating as a separable solid [33][40]. For Cr^{3+}, lime/MgO precipitation (1,500–2,500 mg/L) plus biological treatment (Citrobacter freundii) achieved Cr^{3+} <2.0 mg/L [48][49]. Hg^{2+}–thiourea showed exceptional selectivity (separation coefficient >7,900 for Hg^{2+}/Zn^{2+}) [35].

Comparative Advantage: Complexometric titration costs 1–5% of AAS instrumentation, requires no compressed gases, and offers superior selectivity over conventional precipitation, though it is less sensitive at trace concentrations (<0.1 mg/L) [17][46].

7.3 Limitations of the Present Study

Detection Sensitivity: Direct titration is unreliable below 0.5–1.0 mg/L for Pb^{2+}, Cu^{2+}, and Mn^{2+}. For drinking water compliance (Pb: 0.01 mg/L, Hg: 0.001 mg/L), AAS/ICP-MS or preconcentration is necessary [5][43].
Interference from Competing Ions: Selectivity is compromised in matrices containing multiple metals with similar stability constants. Ca^{2+}/Mg^{2+} interfere with Mn^{2+} determination at pH 10, requiring subtraction by difference [47]. Cu^{2+}/Zn^{2+} interfere with Pb^{2+} determination unless masked with KCN [5].
Kinetic Limitations for Inert Metals: Direct EDTA titration of Cr^{3+} is impractical without heating due to slow water exchange ($k_1 \approx 4.4 \times 10^{-6}\ s^{-1}$), requiring back-titration or AAS [36].

Treatment-Scale Limitations: DTC-adsorbent studies were limited to laboratory batch experiments (50–100 mL). Continuous-flow pilot studies and full economic feasibility were not evaluated [38][40].
Geographic Scope: Case studies were drawn from published literature on Kanpur, Chennai, Jajmau, Ludhiana, Mahanadi basin, and East Kolkata Wetlands. No primary sample collection was conducted [11][24][28][43][48][49][50].

7.4 Recommendations and Future Research Directions for India

Standardisation by SPCBs: CPCB and State Pollution Control Boards should publish SOPs for EDTA titration of Pb^{2+}, Cr^{3+}, Cu^{2+}, Mn^{2+}, and Hg^{2+}, including pH buffers, indicators, and masking agents (KCN, triethanolamine, thioglycolic acid) [5]. This would enable consistent monitoring across India's 4,736 NWQMP sites [3].
Integration with National Missions: The complexometric approach should serve as a first-line screening tool under Namami Gange and the National Water Quality Monitoring Programme. Samples exceeding thresholds may be referred for confirmatory AAS/ICP-OES [3][5].

Training and Capacity Building: MoEFCC, in collaboration with CSIR laboratories (NEERI, CLRI, IICT), should train SPCB technicians in complexometric titration, endpoint detection, and interference management.
Low-Cost DTC Adsorbents: Research should graft DTC onto Indian biomass substrates (coconut coir, rice husk, sugarcane bagasse, neem leaf) for affordable Pb^{2+}/Cu^{2+} removal. Pilot column studies (5–20 L/h) should be conducted at industrial sites in Ludhiana, Faridabad, and Aurangabad [38][40].

EDTA-Membrane Systems: Combined EDTA chelation with nanofiltration/RO should be evaluated to achieve drinking-water-quality effluents, including economic assessment of membrane replacement, energy, and EDTA regeneration [17].
Chromium Recovery in Tanneries: Building on the Jajmau pilot study (>99% Cr removal from ~5,010 mg/L raw effluent), research should optimize precipitation-recycling to produce high-quality recovered chromium hydroxide for re-tanning, supported by LCA and TEA [49].

Green Chelating Agents: Given EDTA's poor biodegradability and DTC toxicity concerns, explore renewable alternatives (IDS, PASP, GLDA, plant polyphenols) for stability constants, selectivity, production cost, and environmental fate.
Real-Time Sensors: Develop low-cost, solid-state colorimetric sensors using immobilized indicators (Xylenol Orange, PAN, Dithizone) or DTC-functionalized paper strips for field-deployable screening of Pb^{2+}, Cu^{2+}, and Hg^{2+}.

Chapter 8

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